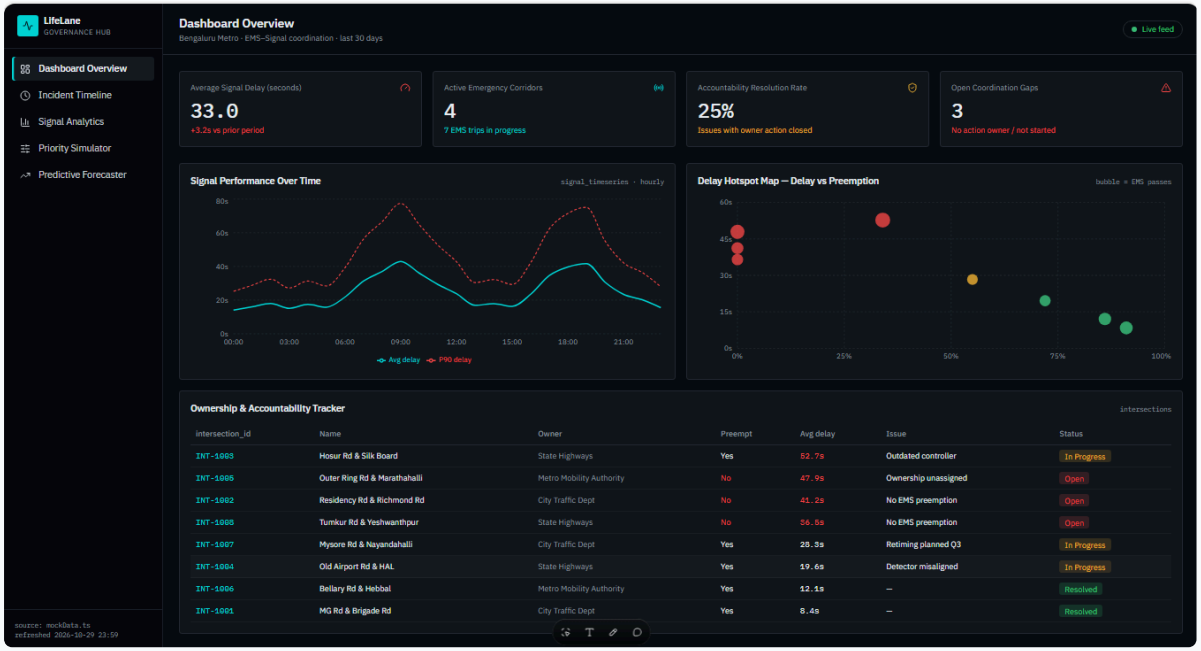
EXECUTIVE DASHBOARD OVERVIEW & ACCOUNTABILITY TRACKER



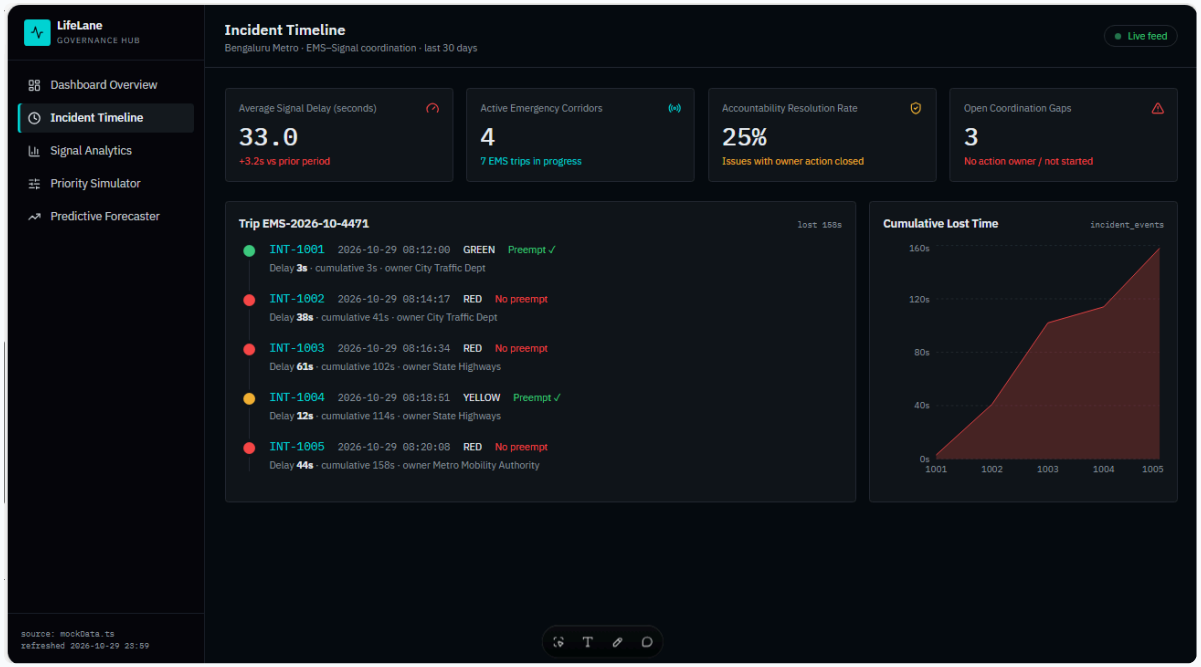
EXECUTIVE METRICS

- **System Baseline:** Average signal delay stands at 33.0 seconds across monitored Bengaluru Metro corridors (+3.2s vs prior period).
- **Active Corridors:** 4 active emergency corridors currently managing 7 simultaneous EMS trips.

GOVERNANCE & ACCOUNTABILITY

- **Resolution Rate:** Only 25% of coordination gaps have closed-loop owner action.
- **Owner Assignment:** Maps intersection IDs (e.g., INT-1003) directly to responsible agencies (State Highways vs. City Traffic Dept).
- **Actionable Fixes:** Flags unassigned owners and outdated controllers to enforce multi-agency accountability.

INCIDENT TIMELINE & CUMULATIVE LOST TIME ANALYSIS



TRIP-LEVEL DIAGNOSTICS

- **Granular Trace:** Replays live emergency runs (e.g., EMS-2026-10-4471) crossing consecutive intersections at exact timestamps.
- **Signal State Capture:** Identifies whether intersections encountered Green, Yellow, or Red phases with preemption status.

IMPACT & RECOMMENDATIONS

- **Cumulative Delay:** Visualizes delay compounding rapidly up to 158 seconds when preemption fails.
- **Agency Tracking:** Automatically attributes uncoordinated delays to specific regional stakeholders.
- **Intervention:** Mandates preemption triggers at high-risk secondary nodes.

SIGNAL ANALYTICS & BOTTLENECK DISTRIBUTION



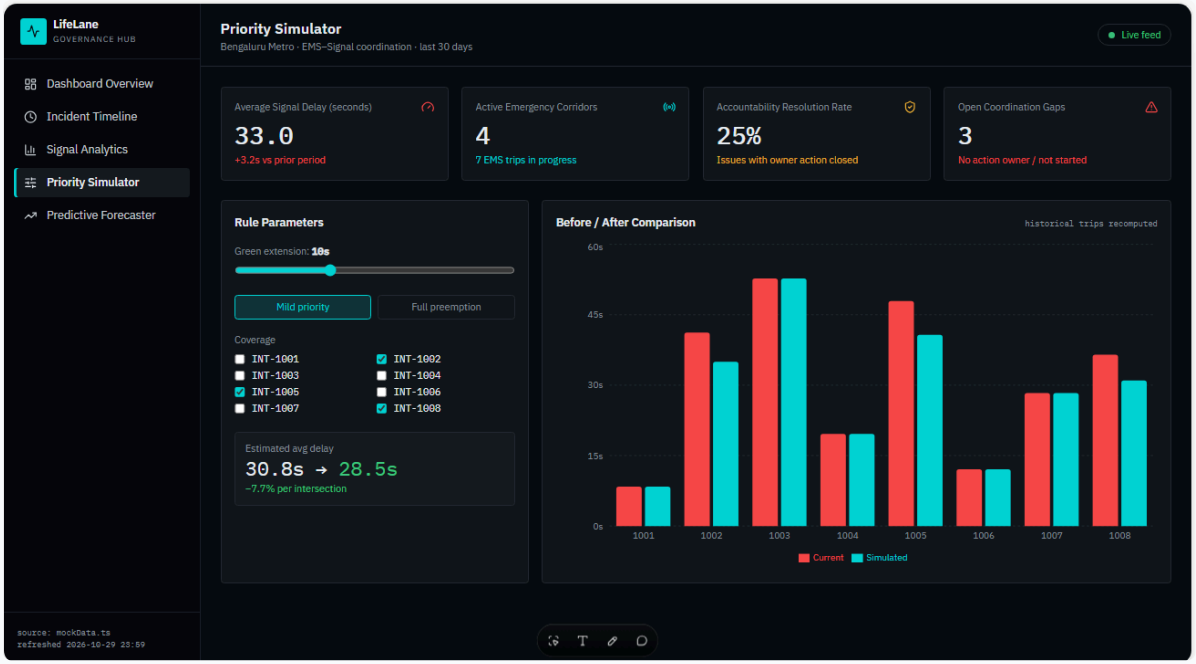
CORRIDOR BOTTLENECKS

- **Intersection Ranking:** Horizontal bar charts isolate top delay offenders exceeding 45–55 second averages.
- **Preemption Rates:** Hourly trigger curves map preemption success against daily traffic density cycles.

STRATEGIC INSIGHTS

- **P90 Variance:** Highlights extreme tail-risk delays during peak morning and evening rushes.
- **Capital Allocation:** Directs infrastructure modernization budgets toward the top 10% worst-performing signals.

PRIORITY SIMULATOR & POLICY IMPACT MODELING



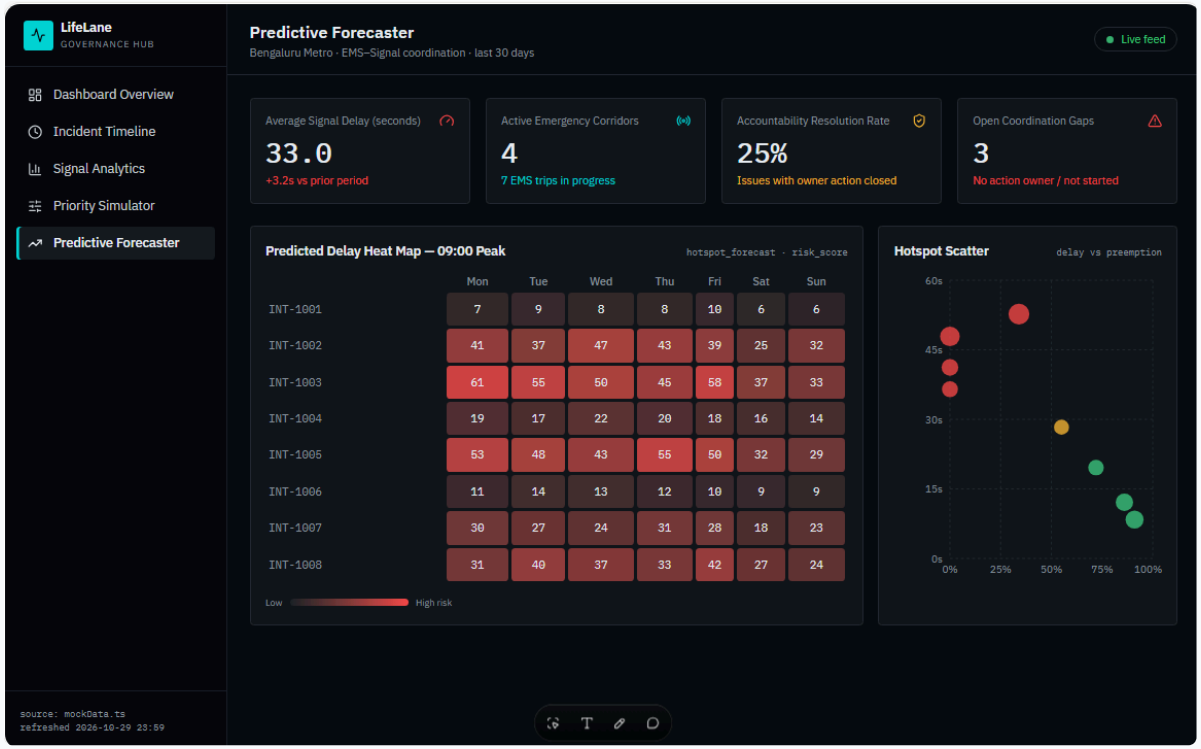
SIMULATION FRAMEWORK

- **Parameter Tuning:** Tests green extension thresholds (10s) and toggles between mild priority and full preemption rules.
- **Targeted Coverage:** Selects specific intersection subsets for localized signal intervention.

QUANTIFIABLE RESULTS

- **Before/After Comparison:** Recomputes historical trips to demonstrate a 7.7% reduction in average delay per intersection.
- **Risk-Free Prototyping:** Allows traffic officials to validate protocol efficacy prior to physical street deployment.

PREDICTIVE FORECASTER & RISK HEATMAP ANALYSIS



PREDICTIVE MODELING

- **Weekly Heat Matrix:** Forecasts congestion severity across days of the week during 09:00 peak hours.
- **Severity Grading:** Color-coded risk scoring identifies vulnerability windows across intersections.

PROACTIVE STRATEGY

- **Resource Scheduling:** Enables emergency dispatchers to reroute ambulances proactively around predicted bottlenecks.
- **Data-Driven Policy:** Moves urban mobility from reactive congestion management to predictive emergency flow.